1. **Implement the numpy basics**

   https://numpy.org/doc/stable/user/basics.html

- Numpy Vs Python's built-in list data structure
  - Memory Consumption: ndarray vs list
  - Time Comparison between Python Lists and Numpy Arrays
- Numpy's array creating functions (eg. .zeros, .ones, .full, .linspace, and so on)
- Different standard numpy data types (eg. bool_, int_, intp, int8, int16, float, complex_, complex64, and so on)

2. **Dataframes with Pandas:**

   https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.html

- Dataframes
  Obtaining Basic Information About DataFrame, Selection and Indexing, Index Basics
- Filtering
  filtering with selection brackets, isin(), between(), contains(), query(), loc[], pandas filter()
- Useful Methods
  apply(), describe(), sort_values(), corr(), idxmin and idxmax, value_counts, replace, unique and nunique, map, duplicated and drop_duplicates, between
- Handling missing data
  isnull(), dropna(), fillna()

3. **Data Visualization**

   The task is to learn the following with Matplotlib (optionally seaborn):
   https://matplotlib.org/stable/users/explain/figure/index.html

- Understanding the Figure Object
- Implementing Figures and Axes
- Figure Parameters
- Subplots Functionality
- Styling - Legends
- Styling - Colors and Styles

You can utilize the data analysis from pandas for plotting visualization.

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4. **FastAPI**
   https://fastapi.tiangolo.com/tutorial/

Implement the following API endpoints for your resource:

- Create: Allow users to add a new resource. This should be accessible via a POST request.
- Read: Allow users to retrieve a list of all resources or a specific resource by its identifier (e.g., via GET requests).
- Update: Allow users to update an existing resource using its identifier. This should be accessible via a PUT request.
- Delete: Allow users to delete a resource by its identifier. This should be accessible via a DELETE request.

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